

Complex Variables

Complex Number: A Complex number is defined by $z = x + iy$, the symbol 'z' is called a complex variable where x, y are real and $i = \sqrt{-1}$. x is the real part and y is the imaginary part.

Modulus and Argument: If $z = x + iy$ then the modulus of z is, $|z| = \sqrt{x^2 + y^2}$ and argument of z is, $\arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$.

Question 1: Find the modulus and argument of $\frac{2+i}{2-i}$.

Solution: Given that, $z = \frac{2+i}{2-i}$

$$\begin{aligned} &= \frac{(2+i)(2+i)}{(2+i)(2-i)} = \frac{(2+i)^2}{2^2 - i^2} \\ &= \frac{3+4i}{4+1} = \frac{3}{5} + \frac{4}{5}i \end{aligned}$$

Modulus of z is, $|z| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9+16}{25}} = 1$

Argument of z is, $= \tan^{-1}\left(\frac{\frac{4}{5}}{\frac{3}{5}}\right) = \tan^{-1}\left(\frac{4}{3}\right) = 53.13^\circ$.

Exercise: Find the modulus and argument of the following function

(i) $z = 2 + i$

(ii) $z = 1 - 3i$

(iii) $z = (1 + i)^2$

(iv) $z = \frac{2-i}{3+i}$

(v) $z = \frac{3+i}{1+i}$

Question: Describe geometrically the region of the following functions,

$$\left| \frac{z-3}{z+3} \right| = 2$$

Solution: Given that, $\left| \frac{z-3}{z+3} \right| = 2$

$$\Rightarrow \frac{|z-3|}{|z+3|} = 2$$

$$\Rightarrow |z-3| = 2|z+3|$$

If, $z = x + iy$, then $|x + iy - 3| = 2|x + iy + 3|$

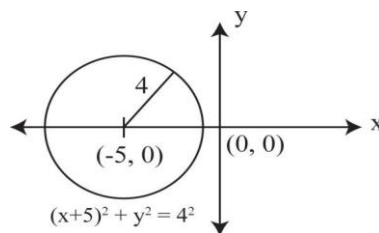
$$\Rightarrow |(x-3) + iy| = 2|(x+3) + iy|$$

$$\Rightarrow \sqrt{(x-3)^2 + y^2} = 2\sqrt{(x+3)^2 + y^2}$$

Squaring both sides, we get,

$$\begin{aligned}(x-3)^2 + y^2 &= 4\{(x+3)^2 + y^2\} \\ \Rightarrow x^2 - 6x + 9 + y^2 &= 4(x^2 + 6x + 9 + y^2) \\ \Rightarrow x^2 - 6x + 9 + y^2 &= 4x^2 + 24x + 36 + 4y^2 \\ \Rightarrow 3x^2 + 30x + 3y^2 + 27 &= 0 \\ \Rightarrow x^2 + 10x + y^2 + 9 &= 0 \\ \Rightarrow x^2 + 2 \cdot x \cdot 5 + 5^2 + y^2 &= 25 - 9 \\ \Rightarrow (x+5)^2 + y^2 &= 16 \\ \therefore (x+5)^2 + y^2 &= 4^2\end{aligned}$$

which represents a circle of radius 4 and centre of $(-5, 0)$



Question-1: Describe geometrically the region of the following functions,

$$|z + 3| + |z - 3| = 10$$

Solution: Given that, $|z + 3| + |z - 3| = 10$

$$\begin{aligned}\Rightarrow |x + iy + 3| + |x + iy - 3| &= 10 \\ \Rightarrow |(x + 3) + iy| + |(x - 3) + iy| &= 10\end{aligned}$$

$$\Rightarrow \sqrt{(x + 3)^2 + y^2} + \sqrt{(x - 3)^2 + y^2} = 10$$

$$\Rightarrow \sqrt{(x + 3)^2 + y^2} = 10 - \sqrt{(x - 3)^2 + y^2}$$

Squaring both sides, we get

$$(x + 3)^2 + y^2 = 100 - 2 \cdot 10 \sqrt{(x - 3)^2 + y^2} + (x - 3)^2 + y^2$$

$$\Rightarrow x^2 + 6x + 9 + y^2 = 100 - 20\sqrt{(x - 3)^2 + y^2} + x^2 - 6x + 9 + y^2$$

$$\Rightarrow 20\sqrt{(x - 3)^2 + y^2} = 100 - 12x$$

$$\Rightarrow 5\sqrt{(x - 3)^2 + y^2} = 25 - 3x$$

Again squaring both sides, we get,

$$\Rightarrow 25\{(x - 3)^2 + y^2\} = (25 - 3x)^2$$

$$\Rightarrow 25(x^2 - 6x + 9 + y^2) = 625 - 150x + 9x^2$$

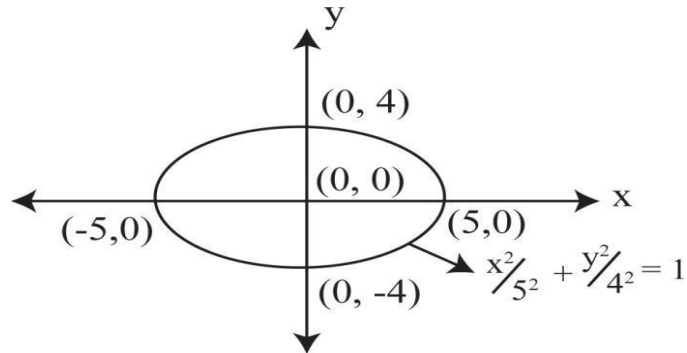
$$\Rightarrow 25x^2 - 150x + 225 + 25y^2 = 625 - 150x + 9x^2$$

$$\Rightarrow 16x^2 + 25y^2 = 400$$

$$\Rightarrow \frac{16}{400}x^2 + \frac{25}{400}y^2 = 1$$

$$\therefore \frac{x^2}{5^2} + \frac{y^2}{4^2} = 1$$

which represents an ellipse passing through the points $(\pm 5, 0)$ and $(0, \pm 4)$.



Question-2: Describe geometrically the region of the following functions:

(i) $\operatorname{Re} \left(\frac{1}{z} \right) \leq 1/2$

(ii) $\operatorname{Im} \left(\frac{1}{z} \right) < 1/2$

(iii) $1 < |z + i| \leq 2$

Solution: (i) Let $z = x + iy$.

Then $\operatorname{Re} \left(\frac{1}{z} \right) \leq \frac{1}{2}$ becomes $\operatorname{Re} \left(\frac{1}{x+iy} \right) \leq \frac{1}{2}$

$$\Rightarrow \operatorname{Re} \left\{ \frac{x-iy}{(x+iy)(x-iy)} \right\} \leq \frac{1}{2}$$

$$\Rightarrow \operatorname{Re} \left(\frac{x-iy}{x^2 + y^2} \right) \leq \frac{1}{2}$$

$$\Rightarrow \operatorname{Re} \left(\frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2} \right) \leq \frac{1}{2}$$

$$\Rightarrow \frac{x}{x^2 + y^2} \leq \frac{1}{2}$$

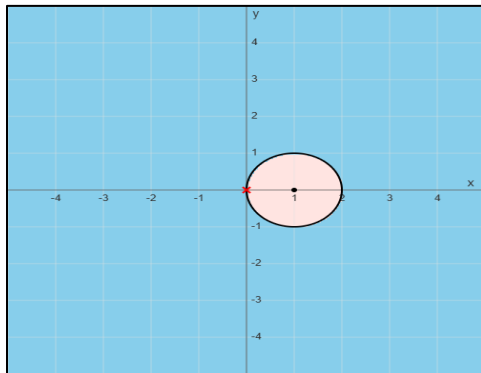
$$\Rightarrow 2x \leq x^2 + y^2$$

$$\Rightarrow x^2 + y^2 \geq 2x$$

$$\Rightarrow x^2 + y^2 - 2x \geq 0$$

$$\Rightarrow (x - 1)^2 + y^2 \geq 1$$

Thus, the region is the set of all external points (including the boundary) of the circle with centre (1,0) and radius 1.



(ii) Given that, $\operatorname{Im} \left(\frac{1}{z} \right) < \frac{1}{2}$

$$\Rightarrow \operatorname{Im} \left(\frac{1}{x + iy} \right) < \frac{1}{2}$$

$$\Rightarrow \operatorname{Im} \left\{ \frac{x - iy}{(x + iy)(x - iy)} \right\} < \frac{1}{2}$$

$$\Rightarrow \operatorname{Im} \left(\frac{x - iy}{x^2 + y^2} \right) < \frac{1}{2}$$

$$\Rightarrow \operatorname{Im} \left(\frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2} \right) < \frac{1}{2}$$

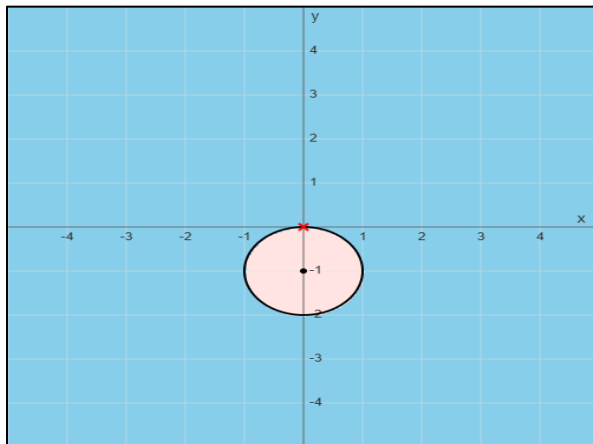
$$\Rightarrow \frac{-y}{x^2 + y^2} < \frac{1}{2}$$

$$\Rightarrow -2y < x^2 + y^2$$

$$\Rightarrow x^2 + y^2 + 2y > 0$$

$$\Rightarrow x^2 + (y + 1)^2 > 1$$

This represents the set of all external points of the circle whose centre is (0,-1) and radius is 1.



(iii) Given that, $1 < |z + i| \leq 2$

$$\Rightarrow 1 < |x + iy + 1| \leq 2$$

$$\Rightarrow 1 < \sqrt{(x^2 + (y + 1)^2)} \leq 2$$

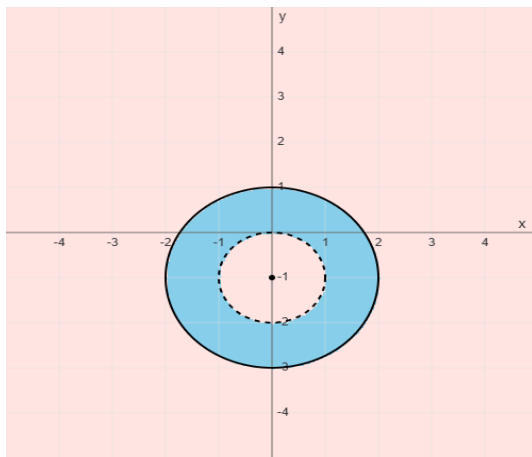
$$\Rightarrow 1 < x^2 + (y+1)^2 \leq 2^2$$

$$\Rightarrow 1 < x^2 + (y+1)^2 \quad \text{and} \quad x^2 + (y+1)^2 \leq 2^2$$

So, $x^2 + (y+1)^2 > 1$ represents all points external to the circle with centre $(0,-1)$ and radius 1.

Also, $x^2 + (y+1)^2 \leq 4$ represents all points internal and boundary of the circle with centre $(0,-1)$ and radius 2.

Thus the region is the set of all common points of external points of the circle $x^2 + (y+1)^2 = 1$ and internal points of the circle $x^2 + (y+1)^2 = 2^2$ including its boundary points.



Exercise: Describe geometrically the region of the following functions

(i) $|z - i| = 2$

(ii) $|z + 2i| + |z - 2i| = 10$

(iii) $|z - 2| = 4|z + 2|$

(iv) $|z + 2| + |z - 2| = 3$

(v) $\left| \frac{z-3}{z+3} \right| = 2.$

$$(vi) \operatorname{Re} \left(\frac{1}{z} \right) < \frac{1}{2}$$

$$(vii) 1 < |z - 2i| \leq 2$$

Analytic Function: A complex function $f(z)$ is said to be analytic at a point z_0 if its derivative exists not only at z_0 but also at each point z in some neighborhood of z_0 .

Entire Function: A complex function $f(z)$ is said to be entire if it is analytic in the whole complex plane.

Simple closed curve: A closed curve which does not intersect itself anywhere is called a simple closed curve.

Contour: A contour is a piecewise smooth path, that is a curve consisting of a finite number of smooth arcs joined end to end.

Cauchy's Integral Formula: Let $f(z)$ be analytic inside and on the boundary of a simply connected closed curve C and a be any point inside C . Then

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)} dz$$

$$\oint_C \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

where C is traversed in the positive sense.

Proof: We know that if $f(z)$ is analytic in a region bounded by two simple closed curves C and C_1 and on these curves then

$$\oint_C f(z) dz = \oint_{C_1} f(z) dz \quad \dots\dots\dots (1)$$

Here the function $\frac{f(z)}{(z-a)}$ is analytic inside and on C except at the point $Z=a$.

$$\oint_C \frac{f(z)}{z-a} dz = \oint_{C_1} \frac{f(z)}{z-a} dz \quad \dots\dots\dots (2)$$

Where C_1 is a circle with center a and radius r .

$$|z - a| = r$$

$$\Rightarrow z - a = re^{i\theta}, 0 \leq \theta \leq 2\pi$$

$$z = a + re^{i\theta}$$

$$dz = 0 + ire^{i\theta} d\theta$$

$$\therefore dz = ire^{i\theta} d\theta$$

Putting those values in (2) we get,

$$\begin{aligned} \oint_c \frac{f(z)}{z - a} dz &= \int_0^{2\pi} \frac{f(a + re^{i\theta})}{re^{i\theta}} ire^{i\theta} d\theta \\ &= i \int_0^{2\pi} f(a + re^{i\theta}) d\theta \end{aligned}$$

Taking limit $r \rightarrow 0$ on both sides and making use of the continuity of $f(z)$ we get,

$$\begin{aligned} \oint_c \frac{f(z)}{z - a} dz &= \lim_{r \rightarrow 0} i \int_0^{2\pi} f(a + re^{i\theta}) d\theta \\ &= if(a) \int_0^{2\pi} d\theta = i \cdot f(a) [\theta]_0^{2\pi} \\ &= i \cdot 2\pi f(a) \\ \Rightarrow f(a) &= \frac{1}{2\pi i} \oint_c \frac{f(z)}{z - a} dz \\ &\text{(Proved)} \end{aligned}$$

Question-1: Show that $\oint_c \frac{e^{3z}}{z - \pi i} dz = -2\pi i$, where C is the circle $|z|=4$.

Solution: Here $f(z) = e^{3z}$ is analytic inside and on the given circle $|z|=4$.

Also $z = a = \pi i$

$$\Rightarrow |z| = |\pi i| = \pi = 3.14 < 4$$

$\therefore z = \pi i$ lies inside the given circle .

Hence by Cauchy's integral formula we have,

$$\begin{aligned}\oint_c \frac{e^{3z}}{z - \pi i} dz &= 2\pi i f(\pi i) \\ &= 2\pi i e^{3\pi i} \\ &= 2\pi i (\cos 3\pi + i \sin 3\pi) \\ &= 2\pi i (-1 + 0) \\ &= -2\pi i.\end{aligned}$$

Question-2: Evaluate $\oint_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $|z|=3$.

Solution: Let $\frac{1}{(z-1)(z-2)} = \frac{A}{z-2} + \frac{B}{z-1} \dots \dots \dots (1)$

$$\Rightarrow 1 = A(z-1) + B(z-2) \dots \dots \dots (2)$$

Putting $z = 2$ and $z = 1$ in (2) we get $A = 1$ and $B = -1$

Therefore $\frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}$

$$\begin{aligned}\text{Hence } \oint_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz \\ = \oint_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)} dz - \oint_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)} dz \dots \dots \dots (3)\end{aligned}$$

Let $f(z) = \sin \pi z^2 + \cos \pi z^2$. Since $z = 1$ and $z = 2$ are inside C and $f(z)$ is analytic inside and on C, so we can apply Cauchy's integral formula.

$$\oint_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)} dz = 2\pi i \{ \sin \pi (1)^2 + \cos \pi (1)^2 \}$$

$$= 2\pi i(\sin\pi + \cos\pi)$$

$$= -2\pi i$$

$$\oint_c \frac{\sin\pi z^2 + \cos\pi z^2}{(z-2)} dz = 2\pi i\{\sin\pi(2)^2 + \cos\pi(2)^2\}$$

$$= 2\pi i$$

Substituting those values in (2), we get

$$\oint_c \frac{\sin\pi z^2 + \cos\pi z^2}{(z-2)} dz = 2\pi i + 2\pi i$$

$$= 4\pi i.$$

Question-3: Show that $\oint_c \frac{e^{tz}}{z^2+1} dz = 2\pi i \sin t$, where C is the circle $|z|=3$ and $t > 0$.

Solution: Here $f(z) = e^{tz}$ is analytic inside and on the given circle $|z|=3$

$$\text{Again } z^2 + 1 = z^2 - (-1) = z^2 - i^2 = (z+i)(z-i)$$

$$\text{Let, } \frac{1}{(z+i)(z-i)} = \frac{A}{z-i} + \frac{B}{z+i}$$

$$\Rightarrow 1 = A(z+i) + B(z-i) \quad \dots \dots \dots (1)$$

Putting $z = i$ and $z = -i$ in (1) we get $A = \frac{1}{2i}$ and $B = -\frac{1}{2i}$

$$\text{Therefore } \frac{1}{z^2+1} = \frac{1}{2i} \left\{ \frac{1}{z-i} - \frac{1}{z+i} \right\}$$

$$\text{Now, } \oint_c \frac{e^{tz}}{z^2+1} dz$$

$$= \frac{1}{2i} \oint_c \frac{e^{tz}}{z-i} dz - \frac{1}{2i} \oint_c \frac{e^{tz}}{z+i} dz \dots \dots \dots (2)$$

Hence by Cauchy's integral formula we get,

$$\oint_c \frac{e^{tz}}{z-i} dz = 2\pi i \times f(i) = 2\pi i e^{it}$$

And

$$\oint_c \frac{e^{tz}}{z+i} dz = 2\pi i \times f(-i) = 2\pi i e^{-it}$$

Substituting these values in (2), we get,

$$\begin{aligned} \oint_c \frac{e^{tz}}{z^2+1} dz &= \frac{1}{2i} (2\pi i e^{it}) - \frac{1}{2i} (2\pi i e^{-it}) \\ &= 2\pi i \left(\frac{e^{it} - e^{-it}}{2i} \right) \end{aligned}$$

$$= 2\pi i \sin t.$$

Exercise:

1. Show that $\oint_c \frac{\sin 3z}{z+\frac{\pi}{2}} dz = 2\pi i$, where C is the circle $|z|=5$.
2. Evaluate $\oint_c \frac{e^{3z}}{z+\pi i} dz$, where C is the circle $|z|=4$.
3. Evaluate $\oint_c \frac{e^{-z}}{z-1} dz$, where C is the circle $|z|=2$.
4. Show that $\oint_c \frac{\sin \pi z + \cos \pi z}{(z-1)(z-2)} dz = 4\pi i$, where C is the circle $|z|=3$.

Harmonic Function: A function which satisfies Laplace equation $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$ in its domain is called Harmonic function.

Conjugate harmonic function: If a function $f(z) = u(x, y) + iv(x, y)$ is analytic in any domain then u and v both are harmonic in that domain and are called conjugate harmonic functions.

Question-1: Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (x^3 - 3xy^2 + 3x^2 - 3y^2 + 1)$$

$$\Rightarrow \frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 6x = \varphi_1(x, y) \text{ (say) } \dots \dots \dots (1)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} (3x^2 - 3y^2 + 6x)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = 6x + 6 \dots \dots \dots (2)$$

$$\text{Again, } \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x^3 - 3xy^2 + 3x^2 - 3y^2 + 1)$$

$$\Rightarrow \frac{\partial u}{\partial y} = -6xy - 6y = \varphi_2(x, y) \text{ (say) } \dots \dots \dots (3)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x} (-6xy - 6y)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = -6x - 6 \dots \dots \dots (4)$$

Adding (2) and (4) we get,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + 6 - 6x - 6 = 0$$

Therefore, u satisfies Laplace equation. So, u is harmonic function.

Let v be the harmonic conjugate of u, so that $f(z) = u + iv$ is analytic.

Putting $x = z, y = 0$ in (1) and (3), we get

$$\varphi_1(z, 0) = 3z^2 + 6z \quad \text{and} \quad \varphi_2(z, 0) = 0$$

By Milne's method we have,

$$f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0)$$

$$= 3z^2 + 6z$$

$$\text{So, } \int f'(z) = \int (3z^2 + 6z)dz$$

$$\Rightarrow f(z) = z^3 + 3z^2 + c_1 + ic_2, \text{ where } c_1 + ic_2 \text{ is a complex constant.}$$

$$\text{Then, } u + iv = (x + iy)^3 + 3(x + iy)^2 + c_1 + ic_2,$$

$$= x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3 + 3(x^2 + 2xiy + i^2y^2) + c_1 + ic_2$$

$$= x^3 + 3x^2iy - 3xy^2 - iy^3 + 3x^2 + 6xiy - 3y^2 + c_1 + ic_2$$

$$= x^3 - 3xy^2 + 3x^2 - 3y^2 + c_1 + i(3x^2y - y^3 + 6xy + c_2)$$

$$\text{Equating imaginary part from both sides we get, } v = 3x^2y - y^3 + 6xy + c_2.$$

Question-2: Show that $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is a harmonic function and hence finds its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = 3x^2y + 2x^2 - y^3 - 2y^2$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(3x^2y + 2x^2 - y^3 - 2y^2)$$

$$\Rightarrow \frac{\partial u}{\partial y} = 6xy + 4x = \varphi_1(x, y) \text{ (say) } \dots \dots \dots (1)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}(6xy + 4x)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = 6y + 4 \dots \dots \dots (2)$$

$$\text{Again, } \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(3x^2y + 2x^2 - y^3 - 2y^2)$$

$$\Rightarrow \frac{\partial u}{\partial y} = 3x^2 - 3y^2 - 4y = \varphi_2(x, y) \text{ (say) } \dots \dots \dots (3)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x}(3x^2 - 3y^2 - 4y)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = -6y - 4 \dots \dots \dots (4)$$

Adding (2) and (4) we get,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6y + 4 - 6y - 4 = 0$$

Therefore, u satisfies Laplace equation. So, u is harmonic function.

Let v be the harmonic conjugate of u, so that $f(z) = u + iv$ is analytic.

Putting $x = z$, $y = 0$ in (1) and (3), we get

$$\varphi_1(z, 0) = 4z \quad \text{and} \quad \varphi_2(z, 0) = 3z^2$$

By Milne's method we have,

$$\begin{aligned} f'(z) &= \varphi_1(z, 0) - i\varphi_2(z, 0) \\ &= 4z - i3z^2 \end{aligned}$$

$$\text{So, } \int f'(z) = \int (4z - i3z^2) dz$$

$$\Rightarrow f(z) = 2z^2 - iz^3 + c_1 + ic_2, \text{ where } c_1 + ic_2 \text{ is a complex constant.}$$

$$\begin{aligned} \text{Then, } u + iv &= 2(x + iy)^2 - (x + iy)^3 + c_1 + ic_2, \\ &= 2(x^2 + 2xiy + i^2y^2) - i(x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) + c_1 + ic_2 \\ &= 2x^2 + 4xiy - 2y^2 - ix^3 + 3x^2y + i3xy^2 - y^3 + c_1 + ic_2 \\ &= 3x^2y + 2x^2 - y^3 - 2y^2 + c_1 + i(4xy - x^3 + 3xy^2 + c_2) \end{aligned}$$

$$\text{Equating imaginary part from both sides we get, } v = 4xy - x^3 + 3xy^2 + c_2.$$

Question-3: Show that $u = x^3 + 6x^2y - 3xy^2 - 2y^3$ is a harmonic function and hence finds its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = x^3 + 6x^2y - 3xy^2 - 2y^3$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (x^3 + 6x^2y - 3xy^2 - 2y^3)$$

$$\Rightarrow \frac{\partial u}{\partial y} = 3x^2 + 12xy - 3y^2 = \varphi_1(x, y) \text{ (say) } \dots \dots \dots (1)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} (3x^2 + 12xy - 3y^2)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = 6x + 12y \dots \dots \dots (2)$$

$$\text{Again, } \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x^3 + 6x^2y - 3xy^2 - 2y^3)$$

$$\Rightarrow \frac{\partial u}{\partial y} = 6x^2 - 6xy - 6y^2 = \varphi_2(x, y) \text{ (say) } \dots \dots \dots (3)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x} (6x^2 - 6xy - 6y^2)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = -6x - 12y \dots \dots \dots (4)$$

Adding (2) and (4) we get,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + 12y - 6x - 12y = 0$$

Therefore, u satisfies Laplace equation. So, u is harmonic function.

Let v be the harmonic conjugate of u, so that $f(z) = u + iv$ is analytic.

Putting $x = z, y = 0$ in (1) and (3), we get

$$\varphi_1(z, 0) = 3z^2 \quad \text{and} \quad \varphi_2(z, 0) = 6z^2$$

By Milne's method we have,

$$f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0)$$

$$= 3z^2 - i6z^2$$

$$= 3(1 - 2i)z^2$$

$$\text{So, } \int f'(z) = \int 3(1 - 2i)z^2 dz$$

$$\Rightarrow f(z) = (1 - 2i)z^3 + c_1 + ic_2, \text{ where } c_1 + ic_2 \text{ is a complex constant.}$$

$$\text{Then, } u + iv = (1 - 2i)(x + iy)^3 + c_1 + ic_2,$$

$$= (x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) - 2i(x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) + c_1 + ic_2$$

$$= x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3 - 2ix^3 - 6x^2i^2y - 6xi^3y^2 - 2i^4y^3 + c_1 + ic_2$$

$$= x^3 + 3x^2iy - 3xy^2 - iy^3 - 2ix^3 + 6x^2y + 6ixy^2 - 2y^3 + c_1 + ic_2$$

$$= x^3 + 6x^2y - 3xy^2 - 2y^3 + c_1 + i(3x^2y - 2x^3 - y^3 + 6xy^2 + c_2)$$

Equating imaginary part from both sides we get, $v = 3x^2y - 2x^3 - y^3 + 6xy^2 + c_2$.

Question-4: Show that $u = e^x(x\cos y - y\sin y)$ is a harmonic function and hence finds its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = e^x(x\cos y - y\sin y)$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} [e^x(x\cos y - y\sin y)]$$

$$\Rightarrow \frac{\partial u}{\partial y} = e^x(x\cos y - y\sin y) + e^x\cos y = \varphi_1(x, y) \text{ (say) } \dots \dots \dots (1)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} [e^x(x\cos y - y\sin y) + e^x\cos y]$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = e^x(x\cos y - y\sin y) + e^x\cos y + e^x\cos y \dots \dots \dots (2)$$

$$\text{Again, } \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} [e^x(x\cos y - y\sin y)]$$

$$\Rightarrow \frac{\partial u}{\partial y} = e^x(-x\sin y - \sin y - y\cos y) = \varphi_2(x, y) \text{ (say) } \dots \dots \dots (3)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x} [e^x(-x \sin y - \sin y - y \cos y)]$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = e^x(-x \cos y - \cos y - \cos y + y \sin y) \dots \dots \dots (4)$$

Adding (2) and (4) we get,

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= e^x(x \cos y - y \sin y) + e^x \cos y + e^x \cos y \\ &\quad + e^x(-x \cos y - \cos y - \cos y + y \sin y) \\ &= e^x(x \cos y - y \sin y + \cos y + \cos y - x \cos y - \cos y - \cos y + y \sin y) \\ &= e^x \cdot 0 = 0 \end{aligned}$$

Therefore, u satisfies Laplace equation. So, u is harmonic function.

Let v be the harmonic conjugate of u, so that $f(z) = u + iv$ is analytic.

Putting $x = z, y = 0$ in (1) and (3), we get

$$\begin{aligned} \varphi_1(z, 0) &= e^z(z \cdot 1 - 0) + e^z \cdot 1 \\ &= ze^z + e^z \end{aligned}$$

$$\text{and } \varphi_2(z, 0) = e^z(-0 - 0 - 0) = 0$$

By Milne's method we have,

$$\begin{aligned} f'(z) &= \varphi_1(z, 0) - i\varphi_2(z, 0) \\ &= ze^z + e^z - i \cdot 0 \end{aligned}$$

$$\text{So, } \int f'(z) = \int (ze^z + e^z) dz$$

$$\Rightarrow f(z) = ze^z - \int 1 \cdot e^z + e^z$$

$$\Rightarrow f(z) = ze^z - e^z + e^z + c_1 + ic_2, \text{ where } c_1 + ic_2 \text{ is a complex constant.}$$

$$\text{Then, } u + iv = ze^z + c_1 + ic_2,$$

$$\begin{aligned}
&= (x + iy)e^{(x+iy)} + c_1 + ic_2 \\
&= (x + iy)e^x \cdot e^{iy} + c_1 + ic_2 \\
&= e^x(x + iy)(\cos y + i \sin y) + c_1 + ic_2 \\
&= e^x(x \cos y + ix \sin y + iy \cos y - y \sin y) + c_1 + ic_2 \\
&= e^x(x \cos y + y \sin y) + ie^x(x \sin y + y \cos y) + c_1 + ic_2
\end{aligned}$$

Equating imaginary part from both sides we get, $v = e^x(x \sin y + y \cos y) + c_2$.

Residues: In Laurent's expansion of any function $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=0}^{\infty} b_n(z - z_0)^{-n}$, coefficient of $(z - z_0)^{-n}$ about the isolated singularity $z = z_0$, is defined as the residue of $f(z)$ at $z = z_0$.

Residue at single pole, $a_{-1} = \lim_{z \rightarrow a} (z - a)f(z)$
Residues at single pole, $a_{-1} = \lim_{z \rightarrow a} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \{(z - a)^m f(z)\}$

Question-1: Find the residues of $f(z) = \frac{e^{3z}}{z + \pi i}$.

Solution: Given that, $f(z) = \frac{e^{3z}}{z + \pi i}$

The poles of $f(z)$ are obtained by solving the equation $z + \pi i = 0$

$$\Rightarrow z = -\pi i$$

$z = -\pi i$, is a single pole.

Residue at $z = -\pi i$ is, $\lim_{z \rightarrow -\pi i} \{(z + \pi i)f(z)\}$

$$= \lim_{z \rightarrow -\pi i} \left\{ (z + \pi i) \frac{e^{3z}}{z + \pi i} \right\}$$

$$= \lim_{z \rightarrow -\pi i} e^{3z} = e^{-\pi i 3} = \cos 3\pi - i \sin 3\pi = (-1) - 0 = -1$$

Question-2: Find the residues of $f(z) = \frac{z^2-2z}{(z+1)^2(z^2+4)}$ at all its poles in a finite plane.

Solution: Given that, $f(z) = \frac{z^2-2z}{(z+1)^2(z^2+4)}$

The poles of $f(z)$ are obtained by solving the equation $(z+1)^2(z^2+4) = 0$

$$\Rightarrow (z+1)^2 = 0 \quad \text{and} \quad (z^2+4) = 0$$

$$\Rightarrow z = -1, -1 \quad \Rightarrow z = \pm 2i$$

$\therefore z = -1$ is a double pole and $z = 2i, z = -2i$ are simple poles.

Residue at $z = -1$ is, $\lim_{z \rightarrow -1} \frac{1}{1!} \frac{d}{dz} \{(z+1)^2 f(z)\}$

$$= \lim_{z \rightarrow -1} \frac{d}{dz} \left\{ (z+1)^2 \frac{z^2-2z}{(z+1)^2(z^2+4)} \right\}$$

$$= \lim_{z \rightarrow -1} \frac{d}{dz} \left\{ \frac{z^2-2z}{(z^2+4)} \right\}$$

$$= \lim_{z \rightarrow -1} \frac{(z^2+4)(2z-2) - (z^2-2z) \cdot 2z}{(z^2+4)^2}$$

$$= \lim_{z \rightarrow -1} \frac{2z^3 + 8z - 2z^2 - 8 - 2z^3 + 4z^2}{(z^2+4)^2}$$

$$= \lim_{z \rightarrow -1} \frac{2z^2 + 8z - 8}{(z^2+4)^2} = \frac{2 - 8 - 8}{(1+4)^2} = \frac{-14}{25}$$

Residue at $z = 2i$ is, $\lim_{z \rightarrow 2i} \{(z-2i)f(z)\}$

$$= \lim_{z \rightarrow 2i} \left\{ (z-2i) \frac{z^2-2z}{(z+1)^2(z+2i)(z-2i)} \right\}$$

$$= \lim_{z \rightarrow 2i} \left\{ \frac{z^2-2z}{(z+1)^2(z+2i)} \right\}$$

$$\begin{aligned}
&= \frac{4i^2 - 4i}{(2i + 1)^2 4i} \\
&= \frac{-4 - 4i}{(4i^2 + 4i + 1) 4i} = \frac{-4 - 4i}{(4i - 3) 4i} = \frac{-1 - i}{4i^2 - 3i} = \frac{-1 - i}{-4 - 3i} \\
&= \frac{1 + i}{4 + 3i} \cdot \frac{4 - 3i}{4 - 3i} = \frac{4 - 3i + 4i - 3i^2}{16 - 9i^2} \\
&= \frac{4 + i + 3}{16 + 9} = \frac{7 + i}{25}
\end{aligned}$$

Residue at $z = -2i$ is, $\lim_{z \rightarrow -2i} \{(z + 2i)f(z)\}$

$$\begin{aligned}
&= \lim_{z \rightarrow -2i} \left\{ (z + 2i) \frac{z^2 - 2z}{(z + 1)^2 (z + 2i)(z - 2i)} \right\} \\
&= \lim_{z \rightarrow -2i} \left\{ \frac{z^2 - 2z}{(z + 1)^2 (z - 2i)} \right\} \\
&= \frac{4i^2 + 4i}{(-2i + 1)^2 (-4i)} \\
&= \frac{-4 + 4i}{(4i^2 - 4i + 1) (-4i)} = \frac{1 - i}{(-4 - 4i + 1) i} = \frac{1 - i}{-4i + 4 + i} = \frac{1 - i}{4 - 3i} \\
&= \frac{1 - i}{4 - 3i} \cdot \frac{4 + 3i}{4 + 3i} = \frac{4 - 4i + 3i - 3i^2}{16 - 9i^2} \\
&= \frac{4 - i + 3}{16 + 9} = \frac{7 - i}{25}.
\end{aligned}$$

Question-3: Find the residues of $f(z) = \frac{e^z}{(z^2 + \pi^2)^2}$ at all its poles in a finite plane.

Solution: Given that, $f(z) = \frac{e^z}{(z^2 + \pi^2)^2} = \frac{e^z}{(z + \pi i)^2 (z - \pi i)^2}$

The poles of $f(z)$ are obtained by solving the equation $(z + \pi i)^2 (z - \pi i)^2 = 0$

$$\Rightarrow (z + \pi i)^2 = 0 \quad \text{and} \quad (z - \pi i)^2 = 0$$

$$\Rightarrow z = -\pi i, -\pi i \quad \Rightarrow z = \pi i, \pi i$$

$\therefore z = \pi i$ and $z = -\pi i$ are two poles of $f(z)$ each of double poles.

Residue at $z = \pi i$ is, $\lim_{z \rightarrow \pi i} \frac{1}{1!} \frac{d}{dz} \{(z - \pi i)^2 f(z)\}$

$$= \lim_{z \rightarrow \pi i} \frac{d}{dz} \left\{ (z - \pi i)^2 \frac{e^z}{(z + \pi i)^2 (z - \pi i)^2} \right\}$$

$$= \lim_{z \rightarrow \pi i} \frac{d}{dz} \left\{ \frac{e^z}{(z + \pi i)^2} \right\}$$

$$= \lim_{z \rightarrow \pi i} \frac{(z + \pi i)^2 e^z - e^z \cdot 2(z + \pi i)}{(z + \pi i)^4}$$

$$= \lim_{z \rightarrow \pi i} \frac{(z + \pi i) \{e^z (z + \pi i) - 2e^z\}}{(z + \pi i)^4}$$

$$= \lim_{z \rightarrow \pi i} \frac{e^z (z + \pi i - 2)}{(z + \pi i)^3} = \frac{e^{\pi i} (2\pi i - 2)}{(2\pi i)^3}$$

$$= \frac{2(\pi i - 1)e^{\pi i}}{-8i\pi^3} = \frac{(\pi i - 1)e^{\pi i}}{-4i\pi^3} = -\frac{(\pi + i)e^{\pi i}}{4\pi^3}$$

Residue at $z = -\pi i$ is, $\lim_{z \rightarrow -\pi i} \frac{1}{1!} \frac{d}{dz} \{(z + \pi i)^2 f(z)\}$

$$= \lim_{z \rightarrow -\pi i} \frac{d}{dz} \left\{ (z + \pi i)^2 \frac{e^z}{(z + \pi i)^2 (z - \pi i)^2} \right\}$$

$$= \lim_{z \rightarrow -\pi i} \frac{d}{dz} \left\{ \frac{e^z}{(z - \pi i)^2} \right\}$$

$$= \lim_{z \rightarrow -\pi i} \frac{(z - \pi i)^2 e^z - e^z \cdot 2(z - \pi i)}{(z - \pi i)^4}$$

$$= \lim_{z \rightarrow -\pi i} \frac{(z - \pi i) \{e^z (z - \pi i) - 2e^z\}}{(z - \pi i)^4}$$

$$\begin{aligned}
&= \lim_{z \rightarrow -\pi i} \frac{e^z (z - \pi i - 2)}{(z + \pi i)^3} = \frac{e^{-\pi i} (-2\pi i - 2)}{(2\pi i)^3} \\
&= \frac{-2(\pi i + 1)e^{-\pi i}}{8i\pi^3} = \frac{-(\pi i - i^2)e^{-\pi i}}{4i\pi^3} = \frac{-(\pi - i)e^{-\pi i}}{4\pi^3}.
\end{aligned}$$

Question-4: Find the residues of $f(z) = \frac{z^2}{z^2 + a^2}$ for only positive value of z .

Solution: Given that, $f(z) = \frac{z^2}{z^2 + a^2}$

The poles of $f(z)$ are obtained by solving the equation $z^2 + a^2 = 0$

$$z = \pm\sqrt{-a^2} = \pm\sqrt{a^2 i^2} = \pm ai$$

The positive value of $z = ai$, which is a single pole.

Residue at $z = ai$ is, $\lim_{z \rightarrow ai} \{(z - ai)f(z)\}$

$$\begin{aligned}
&= \lim_{z \rightarrow ai} \left\{ (z - ai) \frac{z^2}{z^2 + a^2} \right\} \\
&= \lim_{z \rightarrow ai} \left\{ (z - ai) \frac{z^2}{(z + ai)(z - ai)} \right\} \\
&= \lim_{z \rightarrow ai} \left\{ \frac{z^2}{(z + ai)} \right\} \\
&= \frac{(ai)^2}{ai + ai} = \frac{-a^2}{2ai} = \frac{ai^2}{2i} = \frac{ai}{2}.
\end{aligned}$$

Cauchy's Residue theorem: If $f(z)$ is analytic inside and on a simple closed curve C except at a finite number of points $a, b, c, \dots$ inside C at which the residues are $a_{-1}, b_{-1}, c_{-1}, \dots$ respectively, then

$$\oint_C f(z) dz = 2\pi i (a_{-1} + b_{-1} + c_{-1} + \dots)$$

$$= 2\pi i (\text{sum of the residues})$$

Question-1: By use of Cauchy's Residue theorem evaluate $\oint_C \frac{e^{tz}}{(z^2+1)^2} dz$, where C is the circle $|z| = 3$ and $t > 0$.

Solution: Here the circle $|z| = 3$. The poles of $\frac{e^{tz}}{(z^2+1)^2}$ are obtained by solving the equation $(z^2 + 1)^2 = 0$

$$\Rightarrow z^2 + 1 = 0 \Rightarrow z^2 - (-1) = 0$$

$$\Rightarrow z^2 - i^2 = 0$$

$$\Rightarrow (z + i)(z - i) = 0$$

$$\Rightarrow z = i, -i$$

Both poles are double poles and lie inside the circle C, since $|i| = |-i| = 1 < 3$ and $|i| = 1 < 3$.

Residue at $z = i$ is, $\lim_{z \rightarrow i} \frac{1}{1!} \frac{d}{dz} \{(z - i)^2 f(z)\}$

$$= \lim_{z \rightarrow i} \frac{d}{dz} \left\{ (z - i)^2 \frac{e^{tz}}{(z + i)^2 (z - i)^2} \right\}$$

$$= \lim_{z \rightarrow i} \frac{d}{dz} \left\{ \frac{e^{tz}}{(z + i)^2} \right\}$$

$$= \lim_{z \rightarrow i} \frac{(z + i)^2 t e^{tz} - e^{tz} \cdot 2(z + i)}{(z + i)^4}$$

$$= \lim_{z \rightarrow i} \frac{(z + i) \{ t e^{tz} (z + i) - 2 e^{tz} \}}{(z + i)^4}$$

$$= \lim_{z \rightarrow i} \frac{t e^{tz} (z + i) - 2 e^{tz}}{(z + i)^3} = \frac{t e^{ti} (i + i) - 2 e^{ti}}{(i + i)^3}$$

$$= \frac{2 i t e^{ti} - 2 e^{ti}}{-8i} = \frac{2(it - 1)e^{it}}{-8i} = \frac{-(t + i)e^{it}}{4}$$

Similarly, residue at $z = -i$ is $\frac{-(t-i)e^{-it}}{4}$.

Therefore, by Cauchy's residue theorem we have,

$$\oint_C f(z) dz = 2\pi i (\text{sum of the residues})$$

$$= -2\pi i \left[\frac{(t+i)e^{it} + (t-i)e^{-it}}{4} \right]$$

$$= -2\pi i \left[\frac{(t+i)e^{it} + (t-i)e^{-it}}{4} \right]$$

$$= -2\pi i \left[\frac{t(e^{it} + e^{-it}) + i(e^{it} - e^{-it})}{4} \right]$$

$$= -2\pi i \left[\frac{2t \cos t + i \cdot 2i \sin t}{4} \right]$$

$$= -\pi i \left[\frac{4t \cos t + 4i^2 \sin t}{4} \right]$$

$$= -\pi i (t \cos t - \sin t)$$

$$= \pi i (\sin t - t \cos t).$$

Question-2: By use of Cauchy's Residue theorem evaluate $\oint_C \frac{e^{-iz}}{(z+3)(z-i)^2} dz$, where C is the circle $|z-1| = 2$.

Solution: Here the circle $|z-1| = 2$. The poles of $\frac{e^{-iz}}{(z+3)(z-i)^2}$ are obtained by solving the equation $(z+3)(z-i)^2 = 0$

$$\Rightarrow z + 3 = 0 \quad \text{and} \quad (z - i)^2 = 0$$

$$\Rightarrow z = -3 \quad \text{and} \quad z = i, i$$

Since $|i| = 1 < 2$ and $|-3| = 3 > 2$, thus the pole $z = i$ is double pole and lie inside the circle C.

$$\begin{aligned}
& \text{Residue at } z = i \text{ is, } \lim_{z \rightarrow i} \frac{1}{1!} \frac{d}{dz} \{(z - i)^2 f(z)\} \\
&= \lim_{z \rightarrow i} \frac{d}{dz} \left\{ (z - i)^2 \frac{e^{-iz}}{(z + 3)(z - i)^2} \right\} \\
&= \lim_{z \rightarrow i} \frac{d}{dz} \left\{ \frac{e^{-iz}}{(z + 3)} \right\} \\
&= \lim_{z \rightarrow i} \frac{(z + 3)(-ie^{-iz}) - e^{-iz}}{(z + 3)^2} \\
&= \frac{(i + 3)(-ie) - e}{(i + 3)^2} \\
&= \frac{-(1 - 3i - 1)e}{-1 + 6i + 9} = \frac{-3ie}{8 + 6i} = \frac{-3ie(4 - 3i)}{2(4 + 3i)(4 - 3i)} \\
&= \frac{-12ie + 9i^2e}{2(16 + 9)} = \frac{-12ie - 9e}{50}.
\end{aligned}$$

Therefore, by Cauchy's residue theorem we have,

$$\begin{aligned}
\oint_C f(z) dz &= 2\pi i (\text{residues at } z = i) \\
&= 2\pi i \left[\frac{-12ie - 9e}{50} \right] \\
&= -2\pi i \left[\frac{(t + i)e^{it} + (t - i)e^{-it}}{4} \right] \\
&= \frac{-12i^2\pi e - 9i\pi e}{25} \\
&= \frac{(12 - 9i)\pi e}{25}.
\end{aligned}$$

Exercise:

1. By use of Cauchy's Residue theorem show that $\oint_C \frac{e^z}{(z^2 + \pi^2)^2} dz = \frac{i}{\pi}$, where C is the circle $|z| = 4$.
2. By use of Cauchy's Residue theorem evaluate $\oint_C \frac{e^{3z}}{z + \pi i} dz$, where C is the circle $|z+1| = 4$.

Laurent's expansion: If $f(z)$ is analytic inside and on the boundary of the ring shaped region R bounded by two concentric circles c_1 and c_2 with centre at z_0 and respective radii r_1 and r_2 ($r_2 < r_1$), then for all z in R,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=0}^{\infty} b_n (z - z_0)^{-n}$$

Where, $a_n = \frac{1}{2\pi i} \oint_{c_1} \frac{f(w)}{(w - z_0)^{n+1}} dw, n = 0, 1, 2.$

$$b_n = \frac{1}{2\pi i} \oint_{c_2} \frac{f(w)}{(w - z_0)^{-n+1}} dw, n = 1, 2, 3, \dots \dots \dots$$

Question-1: Expand the function $f(z) = \frac{1}{z-3}$ in a Laurent series for the region (i) $|z| < 3$, (ii) $|z| > 3$.

Solution: (i) we express $f(z)$ in a manner so that the binomial expansion is valid for $|z| < 3 \left| \frac{z}{3} \right| < 1$

$$\therefore f(z) = \frac{1}{z-3} = \frac{1}{-3\left(1 - \frac{z}{3}\right)} = -\frac{1}{3}\left(1 - \frac{z}{3}\right)^{-1}$$

$$= -\frac{1}{3}\left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \dots\right)$$

$$= -\frac{1}{3} - \frac{z}{9} - \frac{z^2}{27} - \frac{z^3}{81} - \dots$$

(ii) In this case $|z| > 3 \Rightarrow \frac{|z|}{3} > 1 \Rightarrow \left|\frac{3}{z}\right| < 1$

$$\begin{aligned}\therefore f(z) &= \frac{1}{z-3} = \frac{1}{z\left(1-\frac{3}{z}\right)} = \frac{1}{z} \left(1 - \frac{3}{z}\right)^{-1} \\ &= \frac{1}{z} \left(1 + \frac{3}{z} + \frac{3^2}{z^2} + \frac{3^3}{z^3} + \dots\right) \\ &= \frac{1}{z} + \frac{3}{z^2} + \frac{9}{z^3} + \dots\end{aligned}$$

Question-2: Expand the function $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series for the region (i) $1 < |z| < 3$, (ii) $|z| > 3$.

Solution: Given that, $\frac{1}{(z+1)(z+3)} = \frac{1}{(z+1)(-1+3)} + \frac{1}{(-3+1)(z+3)}$

$$= \frac{1}{2(z+1)} - \frac{1}{2(z+3)}$$

(i) $1 < |z| < 3 \Rightarrow 1 < |z| \quad \text{and} \quad |z| < 3$

$$\Rightarrow \frac{1}{|z|} < 1 \quad \Rightarrow \left|\frac{z}{3}\right| < 1$$

we express $f(z)$ in a manner so that the binomial expansion is valid for $1 < |z| < 3$

$$\Rightarrow \frac{1}{|z|} < 1 \quad \text{and} \quad \Rightarrow \left|\frac{z}{3}\right| < 1$$

$$\therefore f(z) = \frac{1}{2} \frac{1}{z+1} - \frac{1}{2} \frac{1}{(z+3)} = \frac{1}{2z} \cdot \frac{1}{1+\frac{1}{z}} - \frac{1}{6} \cdot \frac{1}{1+\frac{z}{3}}$$

$$= \frac{1}{2z} \left(1 + \frac{1}{z}\right)^{-1} - \frac{1}{6} \left(1 + \frac{z}{3}\right)^{-1}$$

$$= \frac{1}{2z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots\right) - \frac{1}{6} \left(1 - \frac{z}{3} + \frac{z^2}{3^2} - \dots\right)$$

$$= \left(\frac{1}{2z} - \frac{1}{2z^2} + \frac{1}{2z^3} - \dots \right) - \left(\frac{1}{6} - \frac{z}{18} + \frac{z^2}{54} - \dots \right)$$

$$(ii) |z| > 3 \Rightarrow |z| > 1 \quad \text{and} \quad |z| > 3$$

$$\Rightarrow \frac{1}{|z|} < 1 \quad \Rightarrow \left| \frac{3}{z} \right| < 1$$

$$\therefore f(z) = \frac{1}{2} \frac{1}{z+1} - \frac{1}{2} \frac{1}{z+3} = \frac{1}{2z} \cdot \frac{1}{1+\frac{1}{z}} - \frac{1}{2z} \cdot \frac{1}{1+\frac{3}{z}}$$

$$= \frac{1}{2z} \left(1 + \frac{1}{z} \right)^{-1} - \frac{1}{2z} \left(1 + \frac{3}{z} \right)^{-1}$$

$$= \frac{1}{2z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots \right) - \frac{1}{2z} \left(1 - \frac{3}{z} + \frac{9}{z^2} - \frac{27}{z^3} + \dots \right)$$

$$= \left(\frac{1}{2z} - \frac{1}{2z^2} + \frac{1}{2z^3} - \frac{1}{2z^4} + \dots \right) - \left(\frac{1}{2z} - \frac{3}{2z^2} + \frac{9}{2z^3} - \frac{27}{2z^4} + \dots \right)$$

$$= \left(\frac{3}{2z^2} - \frac{1}{2z^2} \right) + \left(\frac{1}{2z^3} - \frac{9}{2z^3} \right) + \left(-\frac{1}{2z^4} + \frac{27}{2z^4} \right) + \dots$$

$$= \frac{1}{z^2} - \frac{4}{z^3} + \frac{13}{z^4} - \dots$$

Question-2: Expand the function $f(z) = \frac{z^2}{(z-1)(z-2)}$ in a Laurent series for the region (i) $1 < |z| < 2$, (ii) $0 < |z| < 1$.

Solution: Here, $f(z) = \frac{z^2}{(z-1)(z-2)}$

$$\text{Let, } \frac{z^2}{(z-1)(z-2)} = 1 + \frac{A}{z-1} + \frac{B}{z-2} \dots \dots \dots (1)$$

$$\Rightarrow z^2 = (z-1)(z-2) + A(z-2) + B(z-1) \dots \dots \dots (2)$$

Putting $z = 1$ and $z = 2$ in (2), we get, $A = -1$, $B = 4$

$$\frac{z^2}{(z-1)(z-2)} = 1 - \frac{1}{z-1} + \frac{4}{z-2}$$

$$\therefore f(z) = 1 - \frac{1}{z-1} + \frac{4}{z-2} \dots \dots \dots (3)$$

we expand f(z) in a manner so that the binomial expansion is valid for $1 < |z| < 2$

$$\Rightarrow \frac{1}{|z|} < 1 \quad \text{and} \quad \Rightarrow \left| \frac{z}{2} \right| < 1$$

$$\therefore f(z) = 1 - \frac{1}{z-1} + \frac{4}{z-2}$$

$$= 1 - \frac{1}{z} \cdot \frac{1}{1 - \frac{1}{z}} - \frac{1}{2} \cdot \frac{4}{1 - \frac{z}{2}}$$

$$= 1 - \frac{1}{z} \left(1 - \frac{1}{z} \right)^{-1} - \frac{4}{2} \left(1 - \frac{z}{2} \right)^{-1}$$

$$= 1 - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right) - 2 \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} \dots \right)$$

$$= \left(1 - \frac{1}{z} - \frac{1}{z^2} - \frac{1}{z^3} - \dots \right) - 2 \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} \dots \right)$$

$$= \dots - \frac{1}{z^3} - \frac{1}{z^2} - \frac{1}{z} - 1 - z - \frac{z^2}{2} + \frac{z^3}{2^2} - \dots$$

For 2nd part we have $0 < |z| < 1$

$$\Rightarrow |z| < 1 \quad \text{and} \quad |z| < 2$$

$$\Rightarrow |z| < 1 \quad \text{and} \quad \Rightarrow \left| \frac{z}{2} \right| < 1$$

$$\therefore f(z) = 1 - \frac{1}{z-1} + \frac{4}{z-2}$$

$$\begin{aligned}
&= 1 + \frac{1}{1-z} - \frac{1}{2} \cdot \frac{4}{1-\frac{z}{2}} \\
&= 1 + (1-z)^{-1} - \frac{4}{2} \left(1 - \frac{z}{2}\right)^{-1} \\
&= 1 + (1+z+z^2+z^3+\dots) - 2 \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} \dots\right) \\
&= \left(1 - \frac{1}{z} - \frac{1}{z^2} - \frac{1}{z^3} - \dots\right) - 2 \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} \dots\right) \\
&= \frac{z^2}{2} + \frac{3z^3}{4} + \dots
\end{aligned}$$

Question: Expand the function $f(z) = \frac{3z-3}{(2z-1)(z-2)}$ in a Laurent series for the region (i) $|z| < 1$, (ii) $|z| > 1$.

Solution: Given that, $f(z) = \frac{3z-3}{(2z-1)(z-2)}$

$$\begin{aligned}
&= \frac{\frac{3}{2} - 3}{(2z-1)\left(\frac{1}{2} - 2\right)} + \frac{3 \cdot 2 - 3}{(2 \cdot 2 - 1)(z-2)} \\
&= \frac{-3}{-3(2z-1)} + \frac{3}{3(z-2)} = \frac{1}{(2z-1)} + \frac{1}{(z-2)}
\end{aligned}$$

Now for the region $|z| < 1$, we have $|z| < 2 \Rightarrow \left|\frac{z}{2}\right| < 1$

$$\begin{aligned}
\therefore f(z) &= \frac{1}{(2z-1)} + \frac{1}{(z-2)} \\
&= \frac{1}{-(1-2z)} - \frac{1}{2} \cdot \frac{1}{1-\frac{z}{2}} \\
&= -(1-2z)^{-1} - \frac{1}{2} \left(1 - \frac{z}{2}\right)^{-1}
\end{aligned}$$

$$= -(1 + 2z + 4z^2 + 8z^3 + \dots) - \frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} \dots \right)$$

$$= -\frac{3}{2} - \frac{9z}{4} - \frac{33z^2}{8} - \dots$$

Now for the region $|z| > 1 \Rightarrow \left| \frac{1}{z} \right| < 1$

$$\therefore f(z) = \frac{1}{(2z-1)} + \frac{1}{(z-2)}$$

$$= \frac{1}{2z \left(1 - \frac{1}{2z} \right)} + \frac{1}{z} \cdot \frac{1}{1 - \frac{2}{z}}$$

$$= \frac{1}{2z} \left(1 - \frac{1}{2z} \right)^{-1} + \frac{1}{z} \left(1 - \frac{2}{z} \right)^{-1}$$

$$= \frac{1}{2z} \left(1 + \frac{1}{2z} + \frac{1}{2^2 z^2} + \frac{1}{2^3 z^3} + \dots \right) + \frac{1}{z} \left(1 + \frac{2}{z} + \frac{2^2}{z^2} + \frac{2^3}{z^3} + \dots \right)$$

$$= \left(\frac{1}{2} + 1 \right) \frac{1}{z} + \left(\frac{1}{4} + 2 \right) \frac{1}{z^2} + \left(\frac{1}{8} + 4 \right) \frac{1}{z^3} + \dots$$

$$= \frac{3}{2z} + \frac{9}{4z^2} + \frac{33}{8z^3} + \dots$$

Exercise:

1. Expand the function $f(z) = \frac{1}{z(z-1)}$ in a Laurent series for the region
(i) $0 < |z| < 3$, (ii) $|z| > 2$.
2. Expand the function $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ in a Laurent series for the region
(i) $2 < |z| < 3$, (ii) $|z| < 2$.
3. Expand the function $f(z) = \frac{1}{(z^2+1)(z+2)}$ in a Laurent series for the region
(i) $1 < |z| < 2$, (ii) $|z| > 1$.